# Exercise Set 12: Fiber Bundles and Connections

*A fiber bundle is a space that looks locally like a product but can be globally twisted. A connection is a rule for differentiating sections of a bundle it tells you how to transport vectors along curves. Every gauge field in physics (the electromagnetic field, the weak field, the strong force) is a connection on a principal bundle. The curvature of the connection is the field strength. The Dirac monopole the first topological object in physics is a connection on the Hopf bundle $S^3 \to S^2$ whose curvature integrates to a nonzero integer, forcing charge quantization.*

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## Recall of Definitions and Examples

### Fiber bundles

**R1.** Write the definition of a **fiber bundle** $(E, M, F, \pi)$ where:

(a) $E$ is the **total space**

(b) $M$ is the **base space**

(c) $F$ is the **fiber**

(d) $\pi : E \to M$ is the **projection**

What is a **local trivialization**? What condition must the local trivializations satisfy?

**R2.** Write the definition of the **transition functions** $g_{\alpha\beta} : U_\alpha \cap U_\beta \to \mathrm{Aut}(F)$ for a fiber bundle with trivializations $\phi_\alpha$ over $\{U_\alpha\}$. What **cocycle condition** must they satisfy on triple overlaps?

**R3.** Identify the total space, base, fiber, and projection for each:

(a) The tangent bundle $TM \to M$

(b) The cotangent bundle $T^*M \to M$

(c) The trivial bundle $M \times F \to M$

(d) The Möbius band $\to S^1$

(e) The Hopf bundle $S^3 \to S^2$

**R4.** Write the definitions of:

(a) A **vector bundle** (fiber $= \mathbb{R}^k$ or $\mathbb{C}^k$, structure group $GL(k)$)

(b) A **principal $G$-bundle** (fiber $= G$, structure group $G$ acting by left multiplication)

(c) A **section** of a bundle $E \to M$: a smooth map $s : M \to E$ with $\pi \circ s = \mathrm{id}_M$

**R5.** Give the definition of a **trivial bundle**. When is a bundle non-trivial? Give one example of a trivial and one of a non-trivial bundle over $S^1$.

**Connections**

**R6.** Write the definition of a **connection** $\nabla$ on a vector bundle $E \to M$:

$$\nabla : \Gamma(E) \to \Gamma(T^*M \otimes E)$$

satisfying: for $f \in C^\infty(M)$ and $s \in \Gamma(E)$,

(a) Linearity: $\nabla(s_1 + s_2) = \nabla s_1 + \nabla s_2$

(b) Leibniz: $\nabla(fs) = df \otimes s + f\nabla s$

Given a local frame $\{e_i\}$ for $E$, write the **connection 1-form** $A = (A^i{}_j)$ via $\nabla e_j = A^i{}_j \otimes e_i$.

**R7.** Write the definition of the **curvature** $F = F^\nabla$ of a connection $\nabla$ on $E \to M$:

$$F(X, Y)s = \nabla_X \nabla_Y s - \nabla_Y \nabla_X s - \nabla_{[X,Y]} s$$

for $X, Y \in \mathfrak{X}(M)$, $s \in \Gamma(E)$. In terms of the connection 1-form $A$, write:

$$F = dA + A \wedge A.$$

What is $F$ for the trivial connection $A = 0$?

**R8.** Write the **gauge transformation** of $A$ and $F$: under $g : M \to G$ (a gauge transformation),

$$A \to g^{-1}Ag + g^{-1}dg, \qquad F \to g^{-1}Fg.$$

What does this say about the physical observability of $A$ vs $F$?

**R9.** State the **Bianchi identity**:

$$dF + A \wedge F - F \wedge A = 0.$$

For an abelian gauge group ($G = U(1)$), what does this reduce to?

**R10.** Define **parallel transport** of a section along a curve $\gamma : [0, 1] \to M$. What ODE must a parallel-transported section satisfy? Define the **holonomy** of a connection around a closed loop.

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# Theory

**T1. (Connection transformation Nawata HW10 P1.)**

Let $E \to M$ be a rank-$r$ vector bundle with local trivializations over $U_\alpha$ and $U_\beta$. On each open set the connection writes as a matrix-valued 1-form $A_\alpha$ and $A_\beta$ respectively.

Given transition function $g_{\alpha\beta} : U_\alpha \cap U_\beta \to GL(r)$:

(a) Show the connection 1-forms are related by:

$$A_\beta = g_{\alpha\beta}^{-1} A_\alpha g_{\alpha\beta} + g_{\alpha\beta}^{-1} d(g_{\alpha\beta}).$$

*Hint: write $\nabla$ in both frames and use $e_\beta = g_{\alpha\beta} e_\alpha$.*

(b) Show the curvature forms transform as:

$$F_\beta = g_{\alpha\beta}^{-1} F_\alpha g_{\alpha\beta}.$$

(c) Conclude: $A$ is not globally defined (it's a gauge field), but $F$ transforms tensorially. What does this mean physically?

**T2. (Bianchi identity Nawata HW10 P2.)**

Prove the Bianchi identity $dF + A \wedge F - F \wedge A = 0$ by substituting $F = dA + A \wedge A$ and using $d^2 = 0$.

For an abelian connection ($A$ is a $U(1)$-valued 1-form, so $A \wedge A = 0$):

(a) Show $F = dA$ and $dF = 0$.

(b) In electromagnetism, $F$ is the Faraday 2-form and $dF = 0$ is exactly the homogeneous Maxwell equations ($\nabla \cdot \mathbf{B} = 0$ and Faraday's law). The Bianchi identity IS the homogeneous Maxwell equations.

**T3. (Holonomy and curvature.)**

Let $\gamma$ be a small loop bounding a surface $S \subset M$. Show that the holonomy around $\gamma$ is approximately:

$$\mathrm{Hol}_\gamma \approx \exp\left(\int_S F\right).$$

For an abelian connection:

(a) The holonomy is $e^{i \oint_\gamma A} = e^{i \int_S F}$ (by Stokes).

(b) For the electromagnetic field, this is the **Aharonov-Bohm phase**: a charged particle moving around a solenoid acquires a phase $e^{i \oint A}$ even if $F = 0$ outside the solenoid. The phase detects $A$, not just $F$.

(c) This shows that $A$ (the gauge potential) is more fundamental than $F$ (the field strength) in quantum mechanics, even though classically only $F$ is observable.

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# Computation

**C1. (The tangent bundle and the Möbius band.)**

(a) Show $TS^1$ is a trivial bundle: construct an explicit global frame (nowhere-vanishing section). What is the explicit diffeomorphism $TS^1 \cong S^1 \times \mathbb{R}$?

(b) The Möbius band is the line bundle over $S^1$ with transition function $g_{12} = -1$ on the overlap $U_1 \cap U_2$. Show it is non-trivial by showing it has no nowhere-vanishing section.
*Hint: a global section corresponds to choosing $\pm 1$ consistently on both charts show this is impossible with the transition function $-1$.*

(c) The cylinder is the trivial line bundle over $S^1$ (transition function $g_{12} = +1$). Show it has a nowhere-vanishing section.

(d) Compare: the Möbius band and cylinder have the same fiber ($\mathbb{R}$) and same base ($S^1$) but different topology. What invariant distinguishes them?
*(Answer: the first Stiefel-Whitney class, which we will meet in the next set.)*

**C2. (The Hopf fibration $S^3 \to S^2$ Nawata HW10 P4.)**

Define:

$$S^3 = \{(z_0, z_1) \in \mathbb{C}^2 : |z_0|^2 + |z_1|^2 = 1\}, \qquad S^2 \cong \mathbb{CP}^1 = S^3/U(1)$$

where $U(1)$ acts by $(z_0, z_1) \mapsto (\lambda z_0, \lambda z_1)$.

(a) Show $\mathbb{CP}^1 \cong S^2$ via the map:

$$[z_0; z_1] \mapsto (z_0\bar{z}_1 + \bar{z}_0 z_1,\ -i(z_0\bar{z}_1 - \bar{z}_0 z_1),\ |z_0|^2 - |z_1|^2) \in \mathbb{R}^3.$$

Verify the image lies in $S^2$ (the norm is 1).

(b) Define local trivializations over $U_N = S^2 \setminus \{N\}$ and $U_S = S^2 \setminus \{S\}$:

$$\pi^{-1}(U_N) \to U_N \times S^1 :\ (z_0, z_1) \mapsto \left([z_0; z_1],\ \frac{z_0}{|z_0|}\right)$$

$$\pi^{-1}(U_S) \to U_S \times S^1 :\ (z_0, z_1) \mapsto \left([z_0; z_1],\ \frac{z_1}{|z_1|}\right)$$

Find the transition function $g_{NS} : U_N \cap U_S \to U(1)$.
*Hint: on the overlap, $\frac{z_0}{|z_0|} = g_{NS} \cdot \frac{z_1}{|z_1|}$.*

(c) Show the transition function is $g_{NS}([z_0; z_1]) = z_0/|z_0| \cdot |z_1|/z_1 = z_0\bar{z}_1/|z_0 z_1|$. In spherical coordinates $(\theta, \varphi)$ on $S^2$: show $g_{NS} = e^{i\varphi}$.

(d) The mapping degree of $g_{NS} : S^1 \to U(1) \cong S^1$ is 1 (it wraps around once). This is the first Chern number of the bundle. What would $g_{NS} = e^{in\varphi}$ (degree $n$) correspond to?

**C3. (The Dirac connection and its curvature Nawata HW10 P4.)**

Define the connection 1-form on $S^3 \subset \mathbb{C}^2$:

$$A = i\,\mathrm{Im}(\bar{z}_0\, dz_0 + \bar{z}_1\, dz_1) = \frac{i}{2}(\bar{z}_0\, dz_0 - z_0\, d\bar{z}_0 + \bar{z}_1\, dz_1 - z_1\, d\bar{z}_1).$$

(a) Pull back $A$ via the section $f_N : U_N \to \pi^{-1}(U_N)$:

$$f_N(\theta, \varphi) = \left(\cos\tfrac{\theta}{2},\, \sin\tfrac{\theta}{2}\, e^{-i\varphi}\right).$$

Show:

$$f_N^* A = -\frac{i}{2}(1 - \cos\theta)\, d\varphi.$$

(b) Pull back $A$ via the section $f_S : U_S \to \pi^{-1}(U_S)$:

$$f_S(\theta, \varphi) = \left(\cos\tfrac{\theta}{2}\, e^{i\varphi},\, \sin\tfrac{\theta}{2}\right).$$

Show:

$$f_S^* A = \frac{i}{2}(1 + \cos\theta)\, d\varphi.$$

(c) Verify that $f_S^* A - f_N^* A = i\, d\varphi$ on $U_N \cap U_S$. This is the gauge transformation by $g_{NS} = e^{i\varphi}$:

$$A_S = A_N + g_{NS}^{-1}\, dg_{NS} = A_N + (-i\, d\varphi) \cdot (-i).$$

So the two local expressions are gauge-equivalent the connection is well-defined on all of $S^2$.

(d) Compute the curvature $F = d(f_N^* A)$ on $U_N$:

$$F = d\left(-\frac{i}{2}(1 - \cos\theta)\, d\varphi\right) = ?$$

Show $F = -\frac{i}{2}\sin\theta\, d\theta \wedge d\varphi$ on both patches.

(e) Compute $\frac{1}{2\pi i}\int_{S^2} F$. Show the result is an integer. What integer?
*This integer is the* ***first Chern number*** $c_1$ *of the Hopf bundle.*

**C4. ($SO(3)$ as an $S^1$-bundle over $S^2$ Nawata HW10 P5.)**

(a) Consider the $SO(2)$ subgroup of $SO(3)$: rotations about the $z$-axis. The quotient $SO(3)/SO(2)$ acts transitively on $S^2$ (any two points are related by a rotation). Show $S^2 \cong SO(3)/SO(2)$.

(b) This gives $SO(3)$ the structure of an $S^1 = SO(2)$-bundle over $S^2$. Construct local trivializations over $U_N = S^2 \setminus \{N\}$ and $U_S = S^2 \setminus \{S\}$.

(c) Find the transition function on $U_N \cap U_S$ (the equator). Show it is $g_{NS} : S^1 \to SO(2) \cong S^1$ of degree $\pm 2$.

(d) Construct a $U(1)$-connection on this bundle and compute its curvature. Compare with the Hopf bundle curvature.

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# Physics: The Dirac Monopole and Charge Quantization

*In 1931, Dirac showed that the existence of a single magnetic monopole anywhere in the universe would force all electric charges to be integer multiples of a basic unit. His argument, reformulated in modern language, is: a magnetic monopole is a connection on the Hopf bundle $S^3 \to S^2$, and the quantization of charge follows from the first Chern number being an integer. This is the first instance in physics where topology forces a discrete spectrum it predates the modern understanding of topological field theory by fifty years.*

**P1. (The magnetic monopole field.)**

A magnetic monopole of strength $g$ at the origin of $\mathbb{R}^3$ creates a magnetic field:

$$\mathbf{B} = \frac{g}{4\pi}\frac{\vec{r}}{r^3} = \frac{g}{4\pi r^2}\hat{r}.$$

(a) Show $\nabla \cdot \mathbf{B} = 0$ for $\vec{r} \neq 0$. Away from the monopole, the field is source-free.

(b) As a 2-form on $\mathbb{R}^3 \setminus \{0\}$:

$$B = \frac{g}{4\pi}\frac{x\,dy \wedge dz + y\,dz \wedge dx + z\,dx \wedge dy}{r^3}.$$

Show $dB = 0$.

(c) Compute $\int_{S^2_r} B$ for any sphere of radius $r$ surrounding the origin. Show the result is $g$, independent of $r$.
*This is the magnetic version of Gauss's law from Set 9.*

(d) For an ordinary magnetic field ($\nabla \cdot \mathbf{B} = 0$ everywhere including the origin), Stokes' theorem gives $\int_{S^2} B = 0$. Why can't we apply Stokes to the monopole field? What is the obstruction?

**P2. (The monopole as a bundle connection.)**

The problem is that $B$ is a closed 2-form on $\mathbb{R}^3 \setminus \{0\}$ but not exact there is no global $A$ with $dA = B$. This is exactly the cohomology obstruction from Set 9: $B$ represents a nonzero class in $H^2_{dR}(\mathbb{R}^3 \setminus \{0\}) \cong \mathbb{R}$.

However, we can define $A$ *locally* on two patches:

$$A_N = \frac{g}{4\pi}\frac{1-\cos\theta}{\sin\theta}\,d\varphi = -\frac{g}{4\pi}(1-\cos\theta)\,d\varphi \quad \text{on } U_N = S^2 \setminus \{N\}$$
$$A_S = -\frac{g}{4\pi}\frac{1+\cos\theta}{\sin\theta}\,d\varphi = \frac{g}{4\pi}(1+\cos\theta)\,d\varphi \quad \text{on } U_S = S^2 \setminus \{S\}$$

(a) Verify $dA_N = dA_S = B$ on each patch. So $B$ is exact on each patch just not globally.

(b) On the overlap $U_N \cap U_S$ (the equator): show $A_S - A_N = \frac{g}{2\pi}\, d\varphi$. This is a gauge transformation: $A_S = A_N + d\lambda$ where $\lambda = \frac{g}{2\pi}\varphi$.

(c) For the gauge transformation to be a well-defined $U(1)$ gauge transformation, the transition function $e^{i\lambda} = e^{ig\varphi/2\pi}$ must be single-valued as $\varphi$ goes from 0 to $2\pi$. Show this requires:

$$\frac{g}{2\pi} \cdot 2\pi \in 2\pi\mathbb{Z} \implies g \in 2\pi\mathbb{Z}.$$

*This is the* ***Dirac quantization condition****: the monopole charge must be an integer multiple of* $2\pi$.

(d) Compare with the Hopf bundle from C2–C3: the Dirac connection $A_N$ is exactly $\frac{g}{4\pi} \cdot (-2) f_N^* A$ from C3. The monopole IS the Hopf bundle connection.

**P3. (Charge quantization.)**

Now let a particle of electric charge $q$ move in the field of a magnetic monopole. The quantum mechanical wavefunction picks up a phase under the gauge transformation on the equator:

$$\psi \to e^{iq\lambda}\psi = e^{iqg\varphi/2\pi}\psi.$$

(a) For the wavefunction to be single-valued (as $\varphi \to \varphi + 2\pi$), what condition must $qg$ satisfy?

(b) If a monopole of charge $g = 2\pi$ exists, show that all electric charges $q$ must be integers.

(c) More generally: if a monopole of charge $g$ exists, then all electric charges are integer multiples of $2\pi/g$. This is **Dirac's charge quantization**:

$$qg \in 2\pi\mathbb{Z}.$$

Write one sentence explaining why the existence of a single monopole in the universe forces all charges to be quantized.

(d) In the real world, all observed charges are integer multiples of $e/3$ (quark charges). Does this mean a monopole of charge $g = 6\pi/e$ exists? What would we need to observe to confirm this?

**P4. (The first Chern number.)**

The integer $\frac{1}{2\pi i}\int_{S^2} F$ computed in C3(e) is the **first Chern number** $c_1$ of the $U(1)$-bundle.

(a) Show the monopole charge $g$ is related to the first Chern number by:

$$c_1 = \frac{1}{2\pi i}\int_{S^2} F = \frac{g}{2\pi i} \cdot (-i) = \frac{g}{2\pi}.$$

The Dirac quantization condition $g \in 2\pi\mathbb{Z}$ is exactly the condition $c_1 \in \mathbb{Z}$.

(b) For the Hopf bundle ($g = 2\pi$): $c_1 = 1$. For a bundle with transition function $e^{in\varphi}$: $c_1 = n$. Show that two $U(1)$-bundles over $S^2$ are isomorphic iff they have the same Chern number.

*This classifies all $U(1)$-bundles over $S^2$: they are labelled by $\mathbb{Z}$, one for each integer Chern number.*

(c) **The topological chain.** Connect all the pieces:

| Topology | Physics |
|---|---|
| $U(1)$-bundle over $S^2$ | Monopole field configuration |
| Transition function $e^{in\varphi}$ | Monopole charge $g = 2\pi n$ |
| First Chern number $c_1 = n$ | Charge quantization unit |
| $c_1 \in \mathbb{Z}$ | Dirac quantization condition |
| $n = 1$: Hopf bundle | Minimal monopole |

(d) **The running theme: topology constrains physics.** In Set 4, the winding number detected the hole in $\mathbb{R}^2 \setminus \{0\}$. In Set 9, Gauss's law detected the hole in $\mathbb{R}^3 \setminus \{0\}$. Now the first Chern number classifies $U(1)$-bundles over $S^2$ and forces charge quantization.

Each time, an integer-valued topological invariant constrains a physical quantity to be discrete. Write one sentence stating the general principle.

The Dirac monopole is the simplest non-trivial bundle in physics: the Hopf bundle $S^3 \to S^2$ with a $U(1)$ connection. The first Chern number $c_1 = \frac{1}{2\pi i}\int_{S^2} F$ is an integer because of topology and this integer forces electric charge to be quantized.

This is the deepest example in our running theme: topology (the classification of bundles by $\mathbb{Z}$) constrains physics (the spectrum of electric charges) to be discrete. No continuous deformation can change $c_1$ it is a topological invariant.

In the next and final set, we put everything together: symplectic geometry and Hamiltonian mechanics, the natural home of the Kepler problem that has been waiting since Set 6B.***Reflection.*** *A fiber bundle is a twisted product locally trivial, globally non-trivial. A connection is a rule for parallel transport locally the gauge potential $A$, globally well-defined only up to gauge transformation. The curvature $F = dA + A \wedge A$ is globally defined and gauge-covariant.*

*The Dirac monopole is the simplest non-trivial bundle in physics: the Hopf bundle $S^3 \to S^2$ with a $U(1)$ connection. The first Chern number $c_1 = \frac{1}{2\pi i}\int_{S^2} F$ is an integer because of topology and this integer forces electric charge to be quantized.*

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